Data, center, and spread

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 11.2×10

2. $1100 \div 100$

3. Double 11.5

4. Half of 66

5. Find the range of 0, 2, 5, 10.

6. Find the MAD of 2, 5, 5, 8.

7. Find $|-3|$ and describe it as a distance.

8. A prism has right-triangle ends with sides 3, 4, 5 cm and length 6 cm. Find surface area.

9. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

10. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

11. 7×4

12. A box is 7 cm by 4 cm by 3 cm. Find its volume.

13. Find two sets of five different whole numbers with minimum 2, maximum 12, median 5, and mean 6.

14. Choose the statistical question: (A) How many pages are in one named book? (B) How many pages are in each of 12 books? Explain.

15. Make a dot plot for 5, 5, 6, 7, 7, 7, 9. State the total dots and most frequent value.

16. Find histogram frequencies for 4, 5, 6, 6, 7, 8, 9, 9 using bins 4 to under 6, 6 to under 8, and 8 to under 10.

17. Find the mean of 6, 8, 10, 12. Explain it as a fair share.

18. Order 12, 6, 9, 8, 9, 13 and find the median.

19. Four values have mean 10. Three are 7, 9, 11. Find the missing value.

20. Find the range of 5, 7, 9, 12, 15. Which values determine it?

21. Find the five-number summary and IQR of 5, 6, 8, 10, 12, 14, 16. Exclude the median from each half.

22. Find the five-number summary and IQR of 5, 7, 8, 10, 12, 14, 15, 17.

23. Find the mean and MAD of 5, 7, 11, 13. Show all four distances.

CHALLENGE / Explain your plan and check the result

C1. A histogram uses bins 0 to under 5, 5 to under 10, and 10 to under 15. Data are 0, 4, 5, 5, 9, 10, 12, 14. Find the three frequencies and the fraction in the highest bin. Explain where the value 5 belongs and why.

C2. Six game scores are 2, 4, 6, 6, 8, and 10. A scorekeeper accidentally records 10 as 40. Find the original and incorrect means and medians. Which calculation provides a useful warning that one entry may be wrong?